

1.4.2 Hydrofoil section optimization

Improving efficiency

We started the design phase by selecting the NACA 63412, used for our previous foil, as the baseline to develop a custom geometry via specialized digital and computational software tools. Once the planform area and ideal aspect ratio were defined, we calculated the root and tip chord lengths. We used these metrics to find the corresponding Reynolds numbers, which allowed for properly setting up the airfoils within the whole optimization framework. The chosen approach was multipoint, targeting both takeoff and cruise. We set the operating points: for takeoff, a high lift coefficient was required to guarantee lift-off, achieved with flap usage; for the cruise phase, we maintained a target coefficient of lift while minimizing the drag coefficient for angles of attack between 0° and 3° . Finally, we applied a minimum thickness limit of 11.2% to satisfy structural requirements.

We developed two optimized airfoils, one for the root and another for the tip. We performed this optimization task using AIRFOILTOOLS [1] software. The program defines both pressure and suction sides of the airfoil using Bézier curves and uses a swarm-based algorithm to find shapes that hit all of the target operating requirements. Swarm algorithms are inspired by biological systems such as flocks of birds or schools of fish, where simple local interactions among individuals provide an effective global solution for resolving complex problems.

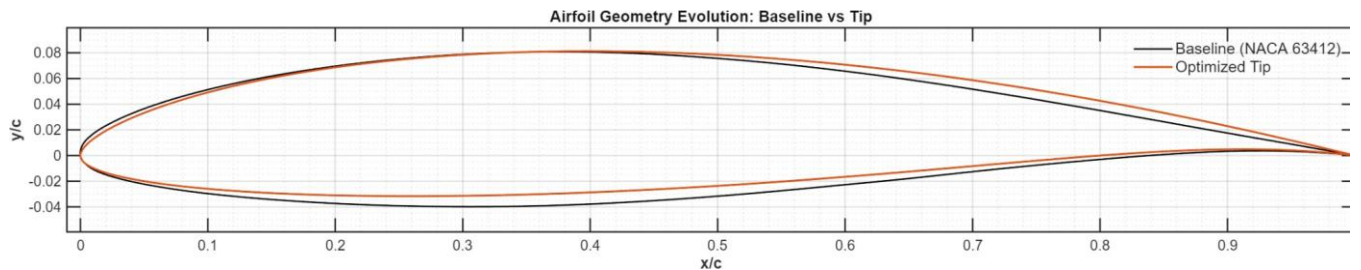


Figure X: Profile overlay of the baseline NACA 63412 and the optimized tip airfoil ($c = 35$ mm).

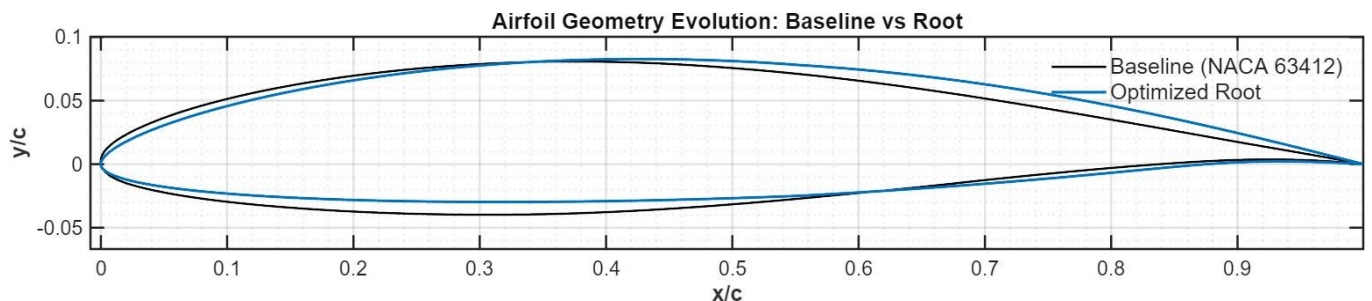


Figure 1: Profile overlay of the baseline NACA 63412 and the optimized root airfoil ($c=100$ mm).

Aerodynamic polars were computed with XFOIL using a critical amplification factor (Ncrit) of 3.5 for both airfoil sections. The tip section was analyzed at a Reynolds number (Re) of 250,000, corresponding to a flight velocity of 14 knots and a chord length of 35 mm. For the root section, the analysis was performed at Re = 710,000 (14 knots and 100 mm chord).

The results demonstrate a significant reduction in the drag coefficient (Cd) within the defined operating window, specifically between 0° and 3° angle of attack (alpha) for cruise conditions.

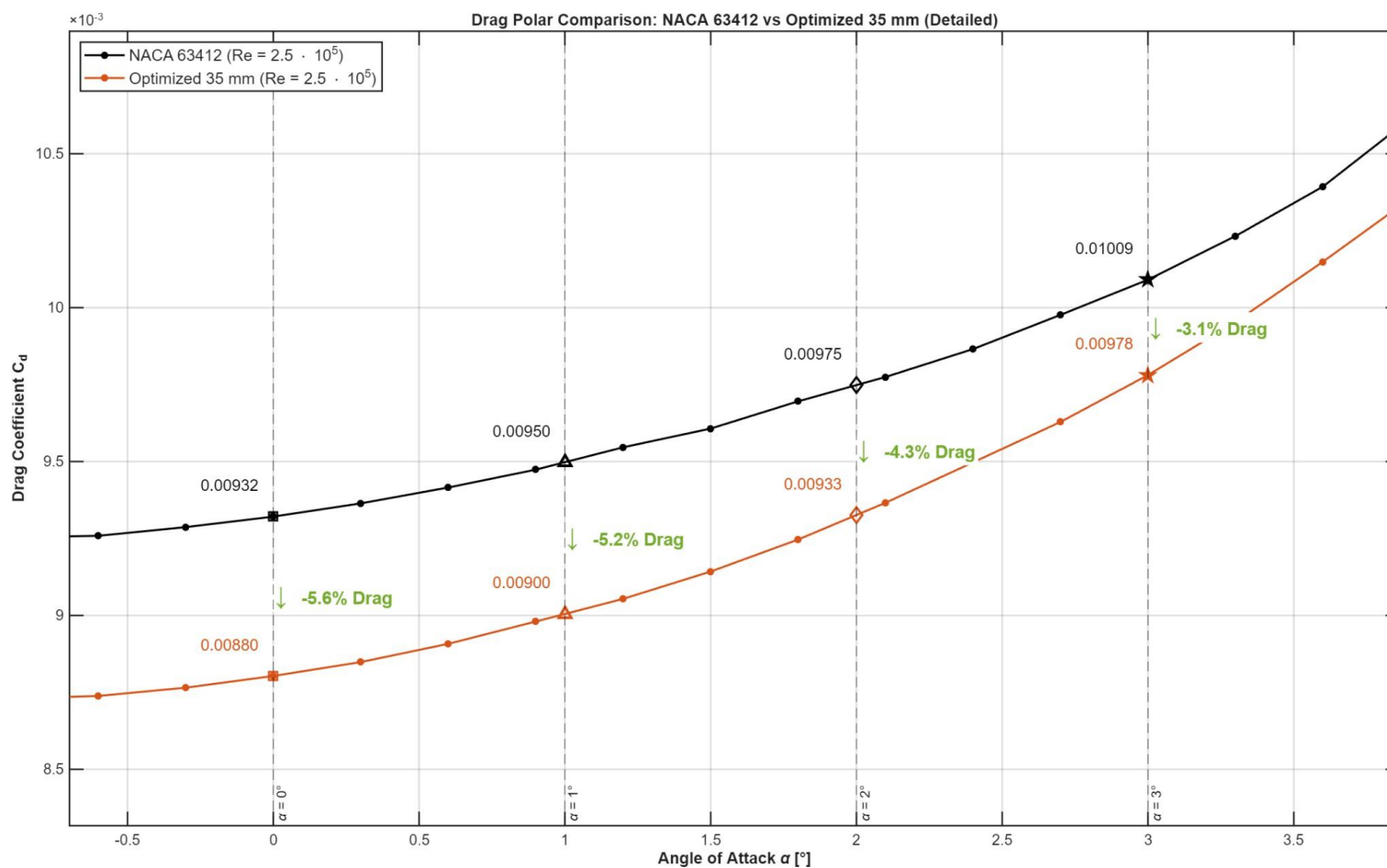


Figure 2: Drag coefficient reduction at the tip section compared to the baseline NACA 63412.

Alpha [°]	Cd NACA 63412	Cd Optimized 35 mm	Reduction [%]
0	0.00932	0.00880	-5.56
1	0.00950	0.00900	-5.19
2	0.00975	0.00933	-4.33
3	0.01009	0.00978	-3.07

Figure x : Drag comparison at Reynolds of 250k

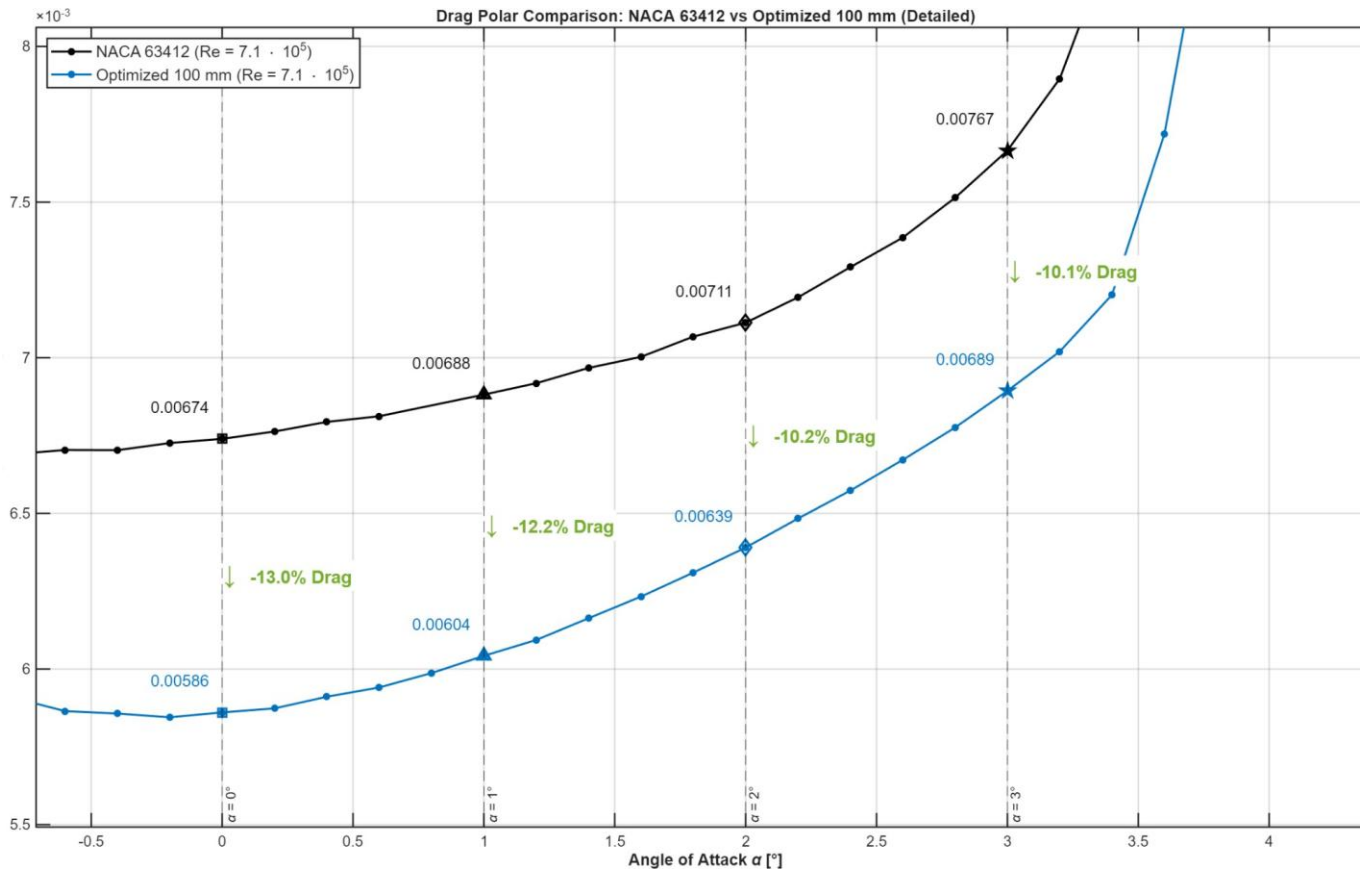


Figure 3: Drag coefficient reduction at the root section compared to the baseline NACA 63412.

Alpha [°]	Cd NACA 63412	Cd Optimized 100 mm	Reduction [%]
0	0.00674	0.00586	-13.04
1	0.00688	0.00604	-12.20
2	0.00711	0.00639	-10.16
3	0.00767	0.00689	-10.07

Figure 3: Drag comparison at Reynolds of 710k

The results reveal a substantial drag reduction, especially regarding the root section at the designated operating points. As is typical in targeted optimization, we observe a trade-off: the drag of our optimized airfoils surpasses the baseline during off-design conditions. We consider this outcome acceptable as such conditions lie beyond the primary flight envelope; specifically, the airfoil is not expected to operate at an angle of attack of 4° or higher during steady-state cruise.

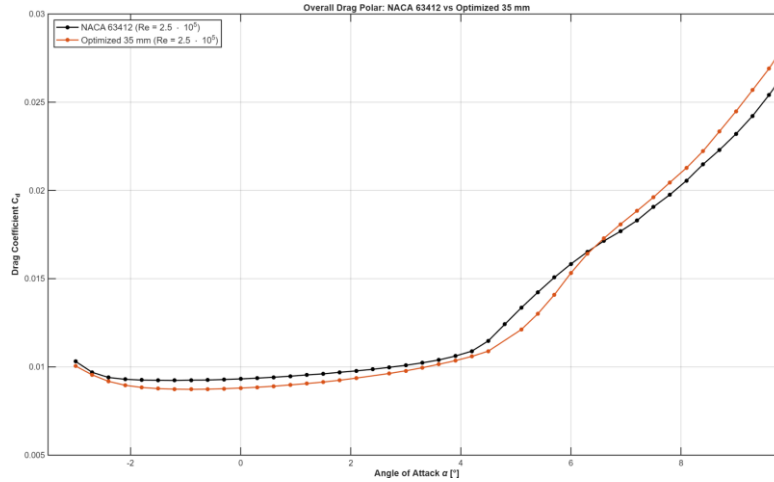


Figure 4: Overall drag of the optimized tip airfoil and the baseline ($Re = 250.000$, $N_{crit} = 3,5$).

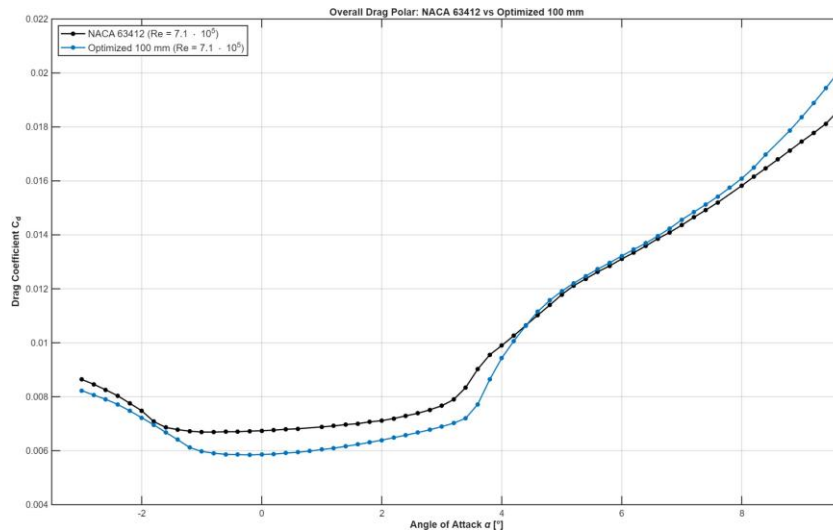


Figure 5: Overall drag of the optimized root airfoil and the baseline ($Re = 710.000$, $N_{crit} = 3,5$).

We have included efficiency plots, which illustrate the ratio of the coefficient of lift to the coefficient of drag, to provide a more comprehensive performance comparison. The results indicate that the optimization successfully shifted the performance peaks to align with the target operating points. Specifically, the maximum efficiency for the tip section increased from 71.1 to 79.6, while the root section saw an improvement from 87.5 to 101.

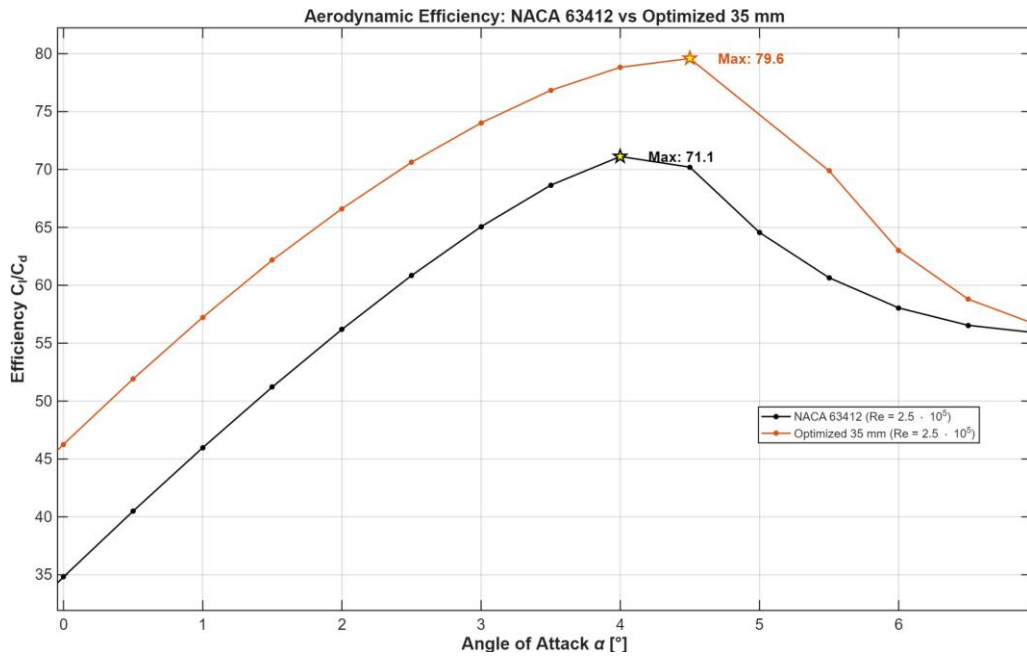


Figure 6: Efficiency (C_l/C_d) plot of baseline and tip optimized ($Re = 710.000$, $N_{crit} = 3,5$).

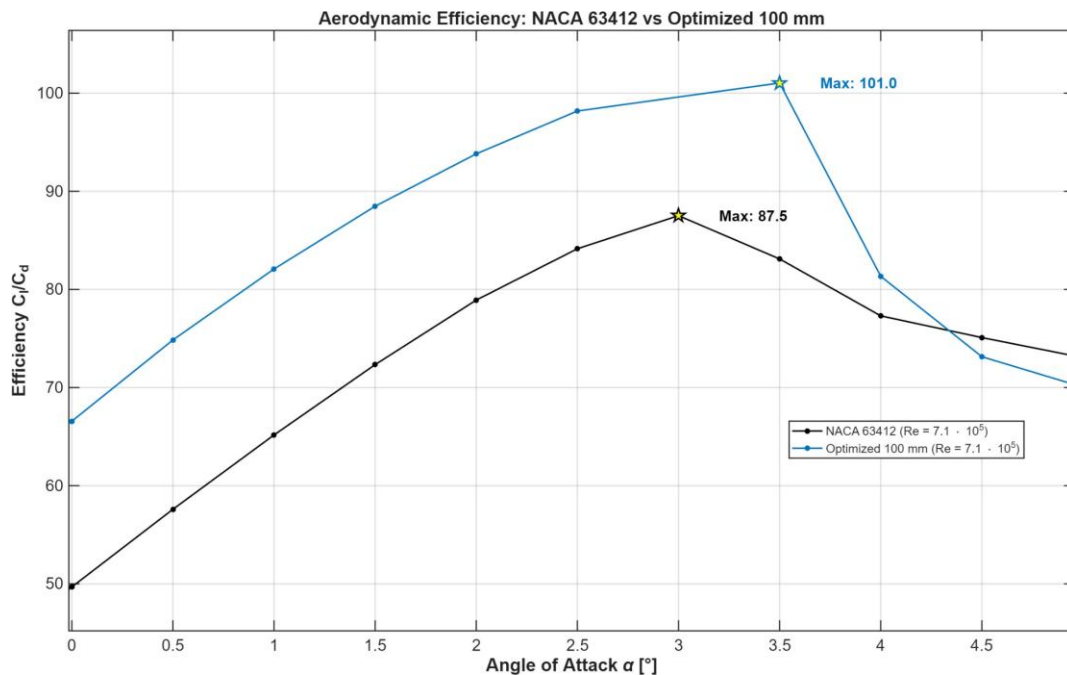
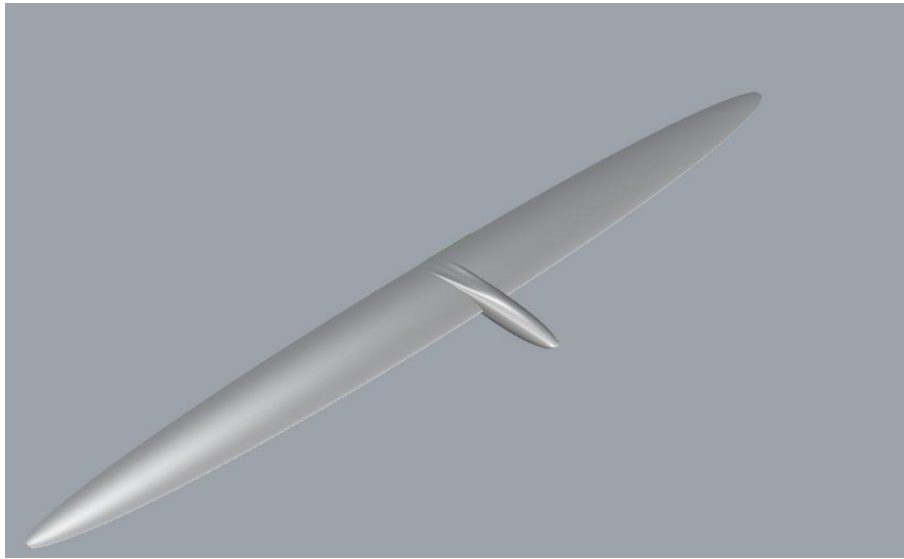


Figure 7: Efficiency (C_l/C_d) plot of baseline and root optimized ($Re = 250.000$, $N_{crit} = 3,5$)

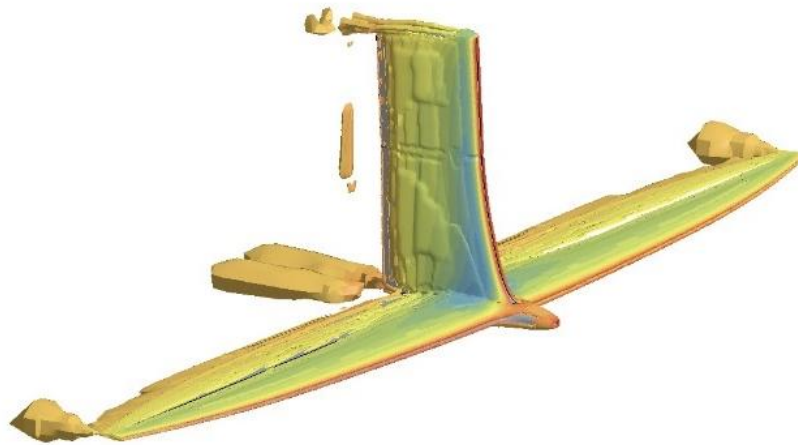
In summary, the multipoint optimization produced two specialized airfoil geometries for the root and tip sections. These profiles successfully achieve the required lift-off performance and a significant drag reduction during cruise, while also strictly adhering to the 11.2% thickness structural constraint to ensure the wing meets all the necessary aerodynamic and loading requirements.

[1] jxjo, *AirfoilEditor* (2025). Available at: <https://github.com/jxjo/AirfoilEditor>. Licensed under the MIT License.



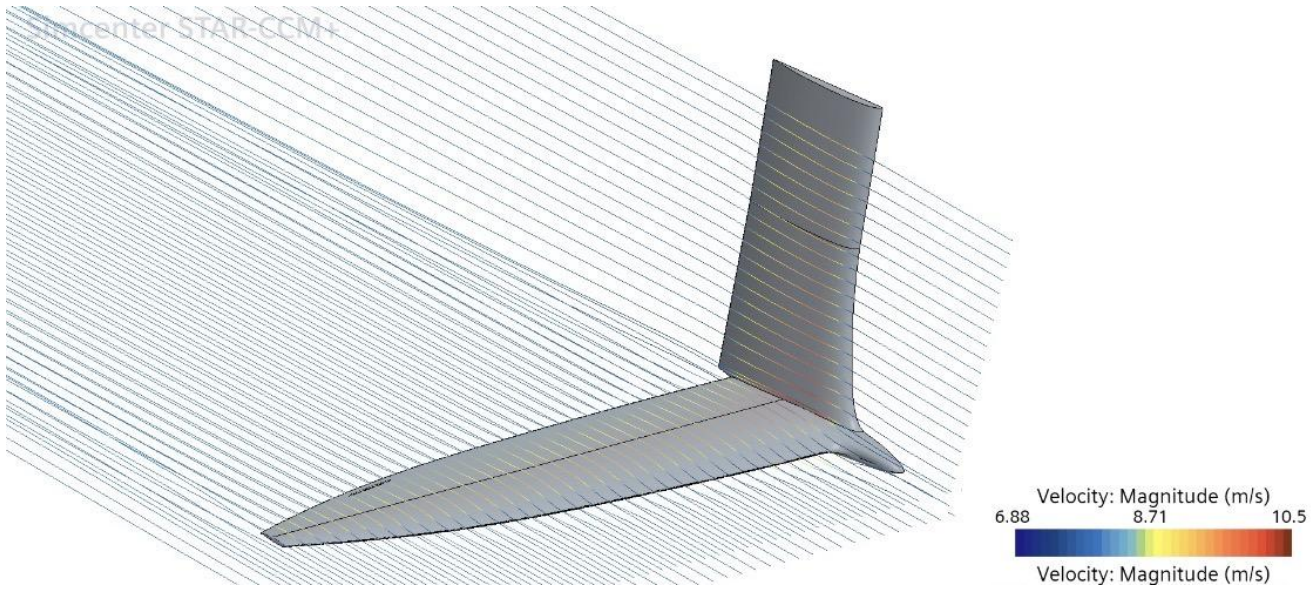
Complete 3D CAD of hydrofoil

Simcenter STAR-CCM+



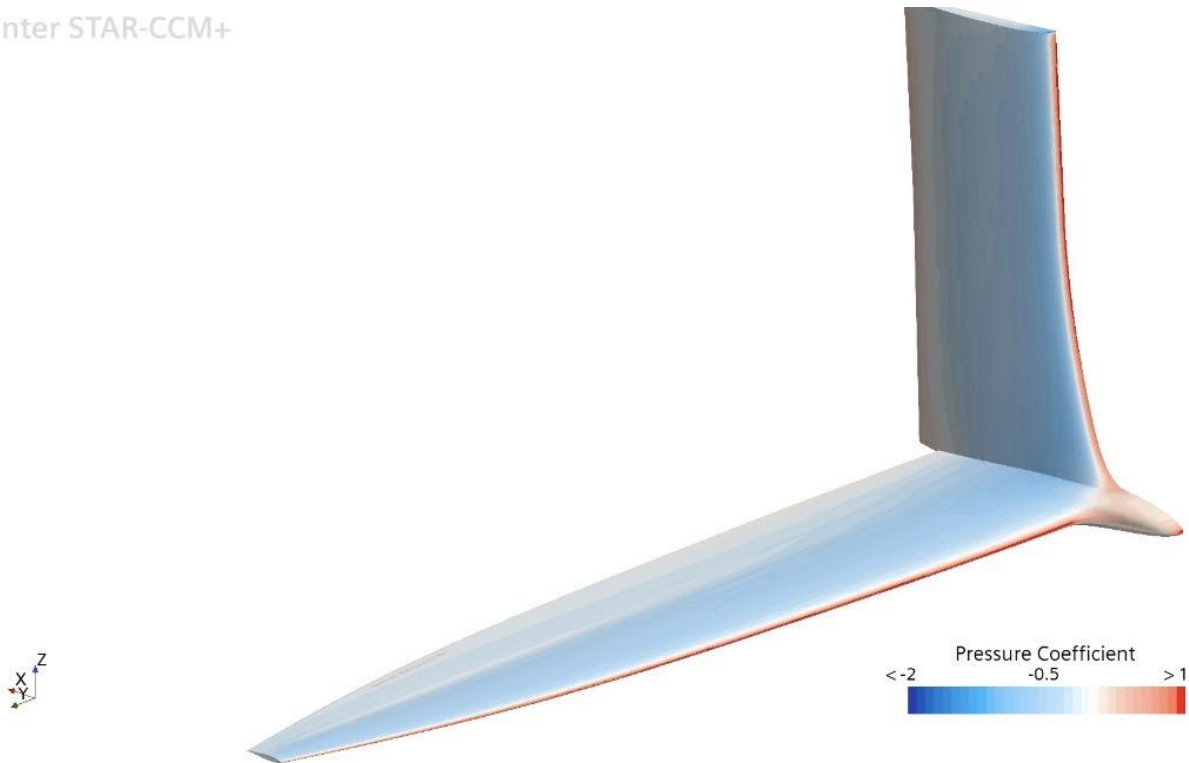
Iso-surface of Q-criterion showing coherent vortex structures and tip vortex core

Simcenter STAR-CCM+



Streamlines

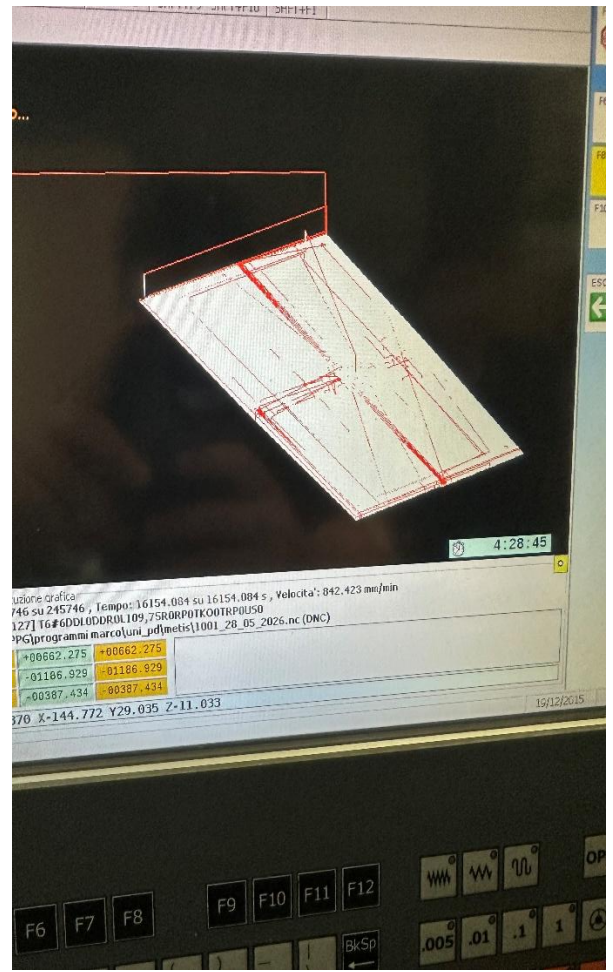
Simcenter STAR-CCM+



Pressure coefficient



Foil mold block preparation



Milling process



Lamination



The foil connected to the vertical strut



The foils in action

The same development process was carried out for the rudder foil as well, but it has been omitted here to avoid redundancy.

Covering the entire workflow, from 2D multi-point optimization and 3D CFD simulations to mold preparation, CNC machining, and carbon-fiber composite lamination, provided an invaluable hands-on engineering experience.

Finally, the sea trials fully validated the computational predictions. The boat exhibited stable flight at a steady cruise speed of 14 knots with a peak of 14.8 knots, successfully matching the design operating envelope and bridging the gap between theoretical modeling and real-world performance.